Pearson Correlation Coefficients (PCC)

https://byjus.com/maths/correlation/
https://towardsdatascience.com/a-step-by-step-implementation-of-principal-component-analysis-5520cc6cd598
https://github.com/AdityaDutt/PCATutorial/blob/main/PCA_tutorial.ipynb
https://github.com/patrickloeber/MLfromscratch/blob/master/mlfromscratch/pca.py
https://kozodoi.me/blog/20230326/pca-from-scratch

Statistics and data science are often concerned about the relationships between two or more variables (or features) of a dataset. Each data point in the dataset is an observation, and the features are the properties or attributes of those observations.

Every dataset you work with uses variables and observations. For example, you might be interested in understanding the following:

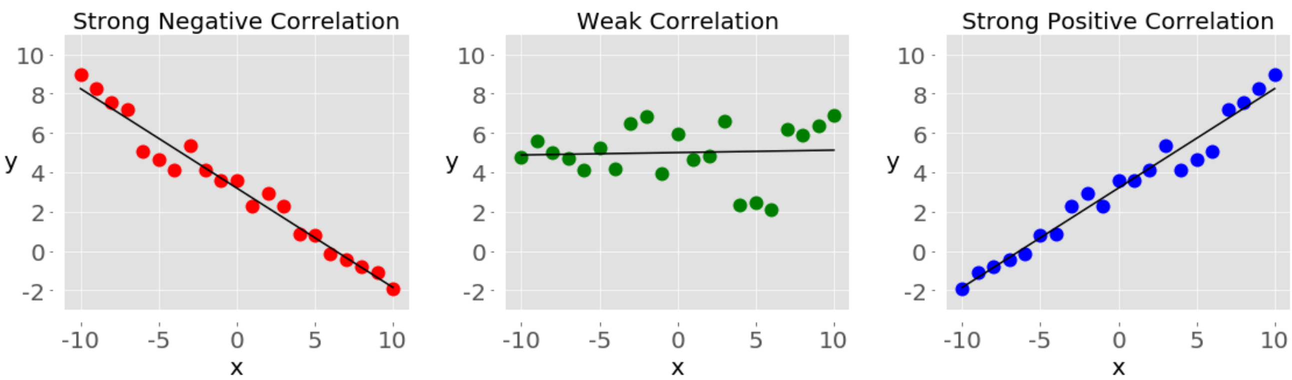
How the height of basketball players is correlated to their shooting accuracy Whether there's a relationship between employee work experience and salary What mathematical dependence exists between the population density and the gross domestic product of different countries In the examples above, the height, shooting accuracy, years of experience, salary, population density, and gross domestic product are the features or variables. The data related to each player, employee, and each country are the observations.

When data is represented in the form of a table, the rows of that table are usually the observations, while the columns are the features. Take a look at this employee table:

Name Years of Experience Annual Salary
Ann 30 120,000
Rob 21 105,000
Tom 19 90,000
Ivy 10 82,000

In this table, each row represents one observation, or the data about one employee (either Ann, Rob, Tom, or Ivy). Each column shows one property or feature (name, experience, or salary) for all the employees.

If you analyze any two features of a dataset, then you'll find some type of correlation between those two features. Consider the following figures:



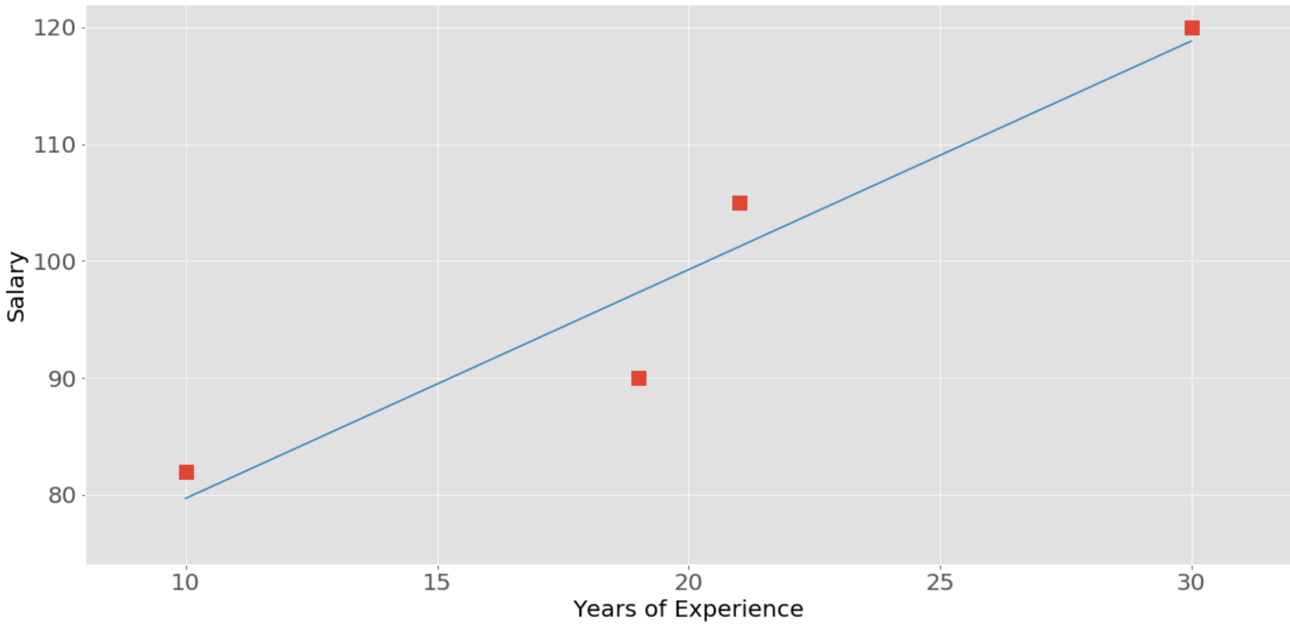
Each of these plots shows one of three different forms of correlation:

Negative correlation (red dots): In the plot on the left, the y values tend to decrease as the x values increase. This shows strong negative correlation, which occurs when large values of one feature correspond to small values of the other, and vice versa.

Weak or no correlation (green dots): The plot in the middle shows no obvious trend. This is a form of weak correlation, which occurs when an association between two features is not obvious or is hardly observable.

Positive correlation (blue dots): In the plot on the right, the y values tend to increase as the x values increase. This illustrates strong positive correlation, which occurs when large values of one feature correspond to large values of the other, and vice versa.

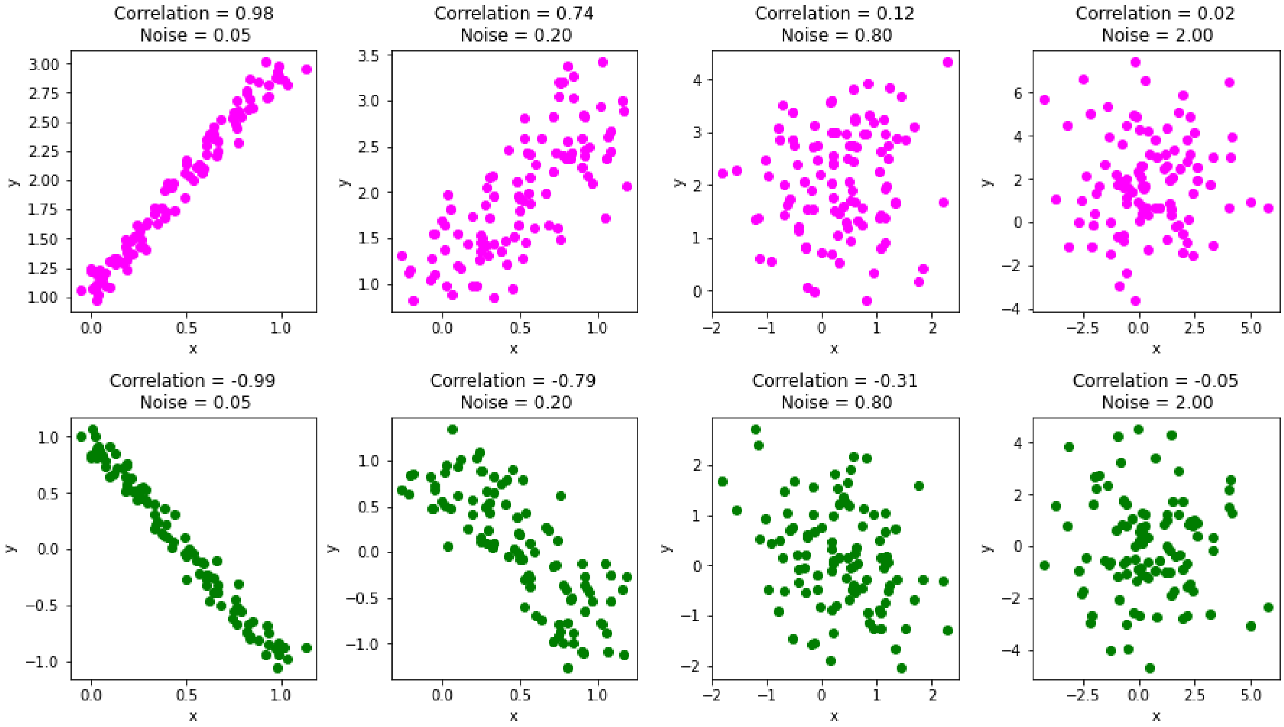
The next figure represents the data from the employee table above:



The correlation between experience and salary is positive because higher experience corresponds to a larger salary and vice versa.

Note: When you’re analyzing correlation, you should always have in mind that correlation does not indicate causation. It quantifies the strength of the relationship between the features of a dataset. Sometimes, the association is caused by a factor common to several features of interest.

Another example of correlation can be seen in the image below:



Step 2: Understanding Correlation

Correlation measures the degree to which two variables move in relation to each other. It's expressed as a value between -1 and 1, known as the correlation coefficient. If the coefficient is close to 1, it signifies a strong positive correlation, while -1 represents a strong negative correlation. A value near 0 indicates little or no relationship.

The correlation coefficient for two variables and can be calculated using the formula:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Where n = Quantity of Information

Σx = Total of the First Variable Value

Σy = Total of the Second Variable Value

Σxy = Sum of the Product of first & Second Value

Σx2 = Sum of the Squares of the First Value

Σy2 = Sum of the Squares of the Second Value

Cálculo del Coeficiente de Correlación de Pearson y Matrices de Covarianza/Correlación

Este cuaderno presenta el cálculo paso a paso del coeficiente de correlación de Pearson y la construcción de las matrices correspondientes, utilizando un conjunto de datos bidimensional de ejemplo.

1. Definición del Problema y Datos

El objetivo principal es normalizar la covarianza para entender la relación lineal entre dos variables X e Y.

Datos de ejemplo:

- X: $[1, 2, 3, 4, 5]$
- Y: $[2, 4, 5, 4, 5]$

2. Cálculo de las Medias

Antes de calcular las desviaciones, determinamos la media poblacional/muestral (μ o $\bar{x}$) para cada variable:

- **Media de X (μ_x):**

$$\mu_x = \frac{1+2+3+4+5}{5} = 3$$

- **Media de Y (μ_y):**

$$\mu_y = \frac{2+4+5+4+5}{5} = 4$$

3. Tabla Completa de Desviaciones y Productos

Para proceder con la covarianza y las varianzas, construimos la siguiente tabla de diferencias respecto a la media, sus productos cruzados y sus cuadrados:

X	Y	$X - \mu_x$	$Y - \mu_y$	$(X - \mu_x)(Y - \mu_y)$	$(X - \mu_x)^2$	$(Y - \mu_y)^2$
1	2	-2	-2	4	4	4
2	4	-1	0	0	1	0
3	5	0	1	0	0	1
4	4	1	0	0	1	0
5	5	2	1	2	4	1
$\sum$				**6**	**10**	**6**

4. Cálculo de Covarianza y Varianzas Muestrales

Utilizando los resultados sumatorios de la tabla anterior y un tamaño de muestra $n = 5$ (con grados de libertad $n - 1 = 4$):

Covarianza ($cov(X, Y)$)

$$\Sigma\text{productos} = 6$$

$$cov(X, Y) = \frac{6}{5-1} = 1.5$$

Varianzas ($Var(X)$ y $Var(Y)$)

- **Varianza de X:**

$$Var(X) = \frac{\sum(X-\mu_x)^2}{n-1} = \frac{10}{4} = 2.5$$

- **Varianza de Y:**

$$Var(Y) = \frac{\sum(Y-\mu_y)^2}{n-1} = \frac{6}{4} = 1.5$$

5. Normalización mediante el Coeficiente de Pearson

Para eliminar la influencia de las escalas, calculamos las desviaciones estándar y finalmente el coeficiente r.

Desviaciones Estándar (σ)

- $\sigma_x = \sqrt{2.5} \approx 1.5811$
- $\sigma_y = \sqrt{1.5} \approx 1.2247$

Coeficiente de Correlación de Pearson (r)

La fórmula de normalización es:

$$r = \frac{cov(X, Y)}{\sigma_X \sigma_Y}$$

Sustituyendo los valores:

$$r = \frac{1.5}{1.5811 \times 1.2247} = \frac{1.5}{1.93649} \approx 0.7746$$

Interpretación: Existe una correlación positiva considerable ($r \approx 0.77$) entre las variables X e Y, indicando una tendencia visual lineal creciente en la nube de puntos.

6. Construcción de Matrices

A partir de los componentes analizados, las estructuras matriciales del sistema se definen formalmente de la siguiente manera:

Matriz de Covarianza

Organiza las varianzas en la diagonal principal y la covarianza cruzada en las restantes de forma simétrica:

$$\Sigma = \begin{bmatrix} Var(X) & cov(X, Y) \\ cov(Y, X) & Var(Y) \end{bmatrix} = \begin{bmatrix} 2.5 & 1.5 \\ 1.5 & 1.5 \end{bmatrix}$$

Matriz de Correlación

Resulta de la conversión de la matriz de covarianza dividiendo cada elemento por el producto de las desviaciones estándar correspondientes:

$$R = \begin{bmatrix} 1 & r \\ r & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.7746 \\ 0.7746 & 1 \end{bmatrix}$$

Implementación

A continuación vamos a implementar el código que realiza los todos cálculos para el Iris dataset comparando permutadamente cada columna de este dataset. Plotearemos cada par de columnas con una gráfica de puntos que mostrará las núvenes de puntos etiquetadas con sus respectivas clases. Al final produciremos un mapa de calor de todas las columnas predictores incluyendo la columna de etiquetas. Las gráficas permitirán el análisis de los datos a partir de métricas objetivas. Al final del cuaderno se propone un análisis similar con otro dataset.

In [1]:
```
use strict;
use warnings;
use Data::Dump qw(dump);
use AI::MXNet qw(nd);
use sml qw(show_plot);
IPerl->load_plugin('Chart::Plotly');
```

Step 1: Fetch the data from a dataset

We obtain data from Iris Dataset, which has 4 columns. We use the function previously defined in the book.

In [3]:
```
my ($dataset, $header) = sml->load_csv('../data/iris.csv');
my ($lookup, $rlookup) = sml->str_column_to_int($dataset, -1);
$dataset = nd->array($dataset);
```

Out[3]:
```
<AI::MXNet::NDArray 150x5 @cpu(0)>
```

In [4]:
```
printf "%d-%s ", $_, $header->[$_] for (0 .. $#$header);
printf "%s\n", dump $lookup, $rlookup;
```

```
0-SepalLengthCm 1-SepalWidthCm 2-PetalLengthCm 3-PetalWidthCm 4-Species (
  { "Iris-setosa" => 0, "Iris-versicolor" => 1, "Iris-virginica" => 2 },
  { "0" => "Iris-setosa", "1" => "Iris-versicolor", "2" => "Iris-virginica" },
)
```

Out[4]: 1

In [5]:
```
print $dataset->slice([0, 5])->asstr;
```

```
[[5.1000 3.5000 1.4000 0.2000      0]
 [4.9000      3 1.4000 0.2000      0]
 [4.7000 3.2000 1.3000 0.2000      0]
 [4.6000 3.1000 1.5000 0.2000      0]
 [     5 3.6000 1.4000 0.2000      0]]
```

Out[5]: 1

In [6]:
```
my $X = $dataset->slice(':', [undef, -1]);
printf "%d-%s ", $_, $header->[$_] for (0 .. $#$header -1);
```

```
print $X->slice([undef, 5])->asstr;
```

```
0-SepalLengthCm 1-SepalWidthCm 2-PetalLengthCm 3-PetalWidthCm [[5.1000 3.5000 1.4000 0.2000]
 [4.9000      3 1.4000 0.2000]
 [4.7000 3.2000 1.3000 0.2000]
 [4.6000 3.1000 1.5000 0.2000]
 [     5 3.6000 1.4000 0.2000]]
```

Out[6]: 1

In [7]:
```
my $y = $dataset->slice(':', -1);
printf "y: %s", $y->asstr;
```

```
y: [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2]
```

Out[7]: 1

In [8]:
```
printf "Mean: %s\n", nd->mean($X, axis=>0)->asstr;
printf "Deviaton: %s\n", nd->std($X, axis=>0)->asstr;
# Mean: [5.84333   3.054 3.75867 1.19867]
# Deviaton: [0.825301 0.432147 1.75853 0.760613]
```

```
Mean: [5.8433 3.0540 3.7587 1.1987]
Deviaton: [0.8253 0.4321 1.7585 0.7606]
```

Out[8]: 1

In [9]:
```
# Now, we take any two columns from the dataset, calculate their correlation we plot
my ($X0, $X1, $X2, $X3) = @{$X->T};
```

Out[9]: <AI::MXNet::NDArray 150 @cpu(0)><AI::MXNet::NDArray 150 @cpu(0)><AI::MXNet::NDArray 150 @cpu(0)><AI::MXNet::NDArray 150 @cpu(0)>

In [10]:
```
printf "Pearson Correlation %s x %s: %s\n", @$header[0, 1],
       nd->corrcoef($X0, y=>$X1)->asstr;
# Pearson Correlation X0 x X1
#   [[ 1.          -0.10936925]
#   [-0.10936925  1.         ]]
```

```
Pearson Correlation SepalLengthCm x SepalWidthCm: [[      1 -0.1094]
 [-0.1094       1]]
```

Out[10]: 1

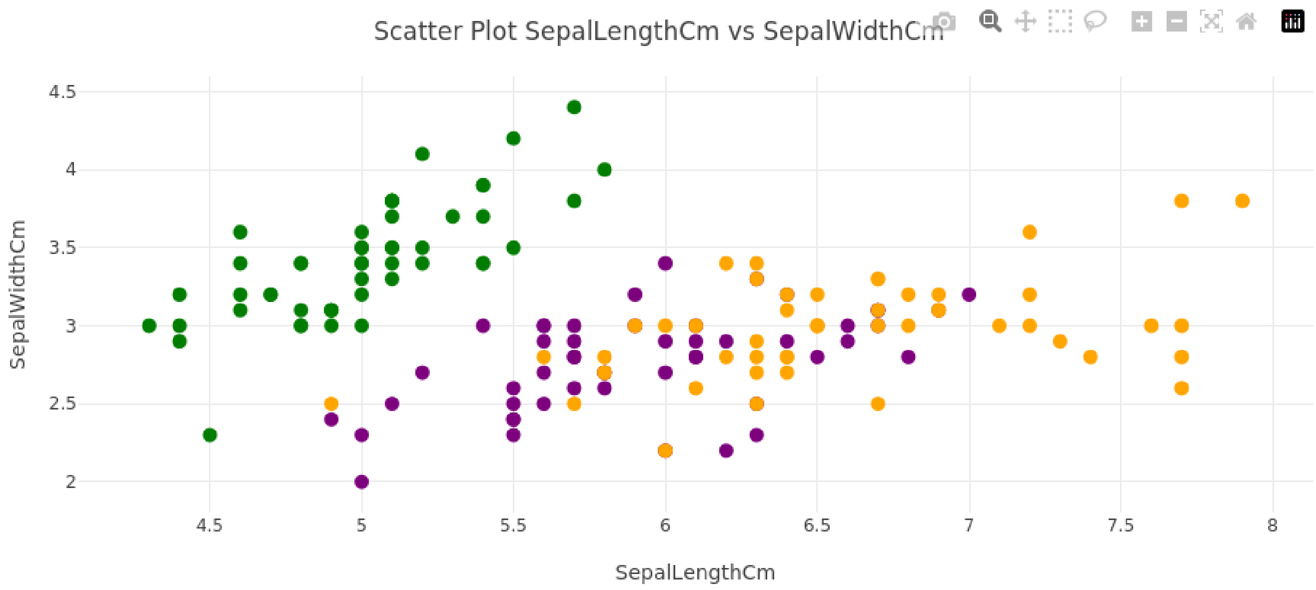
In [11]:
```
my $color_scale = [
    [0,   'green'],
    [0.5, 'purple'],
    [1,   'orange']
];
```

Out[11]: ARRAY(0x5623c209ad70)

In [12]:
```
my $trace = new Chart::Plotly::Trace::Scatter(
    x      => $X0->aspdl,
    y      => $X1->aspdl,
    mode   => 'markers',
    marker => {
        color      => $y->aspdl,
        colorscale => $color_scale,
        cmin       => $y->min,
        cmax       => $y->max,
        size       => 10
    }
);
```

```
my $layout = {title => { text => sprintf('Scatter Plot %s vs %s', @$header[0, 1]) },
              xaxis => { title => $header->[0] }, yaxis => { title => $header->[1] },
              width  => 900, height => 400,
              margin => { l => 50, r => 0, t => 50, b => 50 }};

my $plot = new Chart::Plotly::Plot(traces => [$trace],
                                   layout => $layout);

IPerl->display($plot);
```

In [15]:
```
sml->embed_plot($plot, width=>900, height=>450);#show_plot($plot);
```



In [16]:
```
printf "Pearson Correlation %s x %s: %s\n", @$header[0, 2],
       nd->corrcoef($X0, y=>$X2)->asstr;
# Pearson Correlation X0 x X2
#  [[1.         0.87175416]
#  [0.87175416 1.        ]]
```

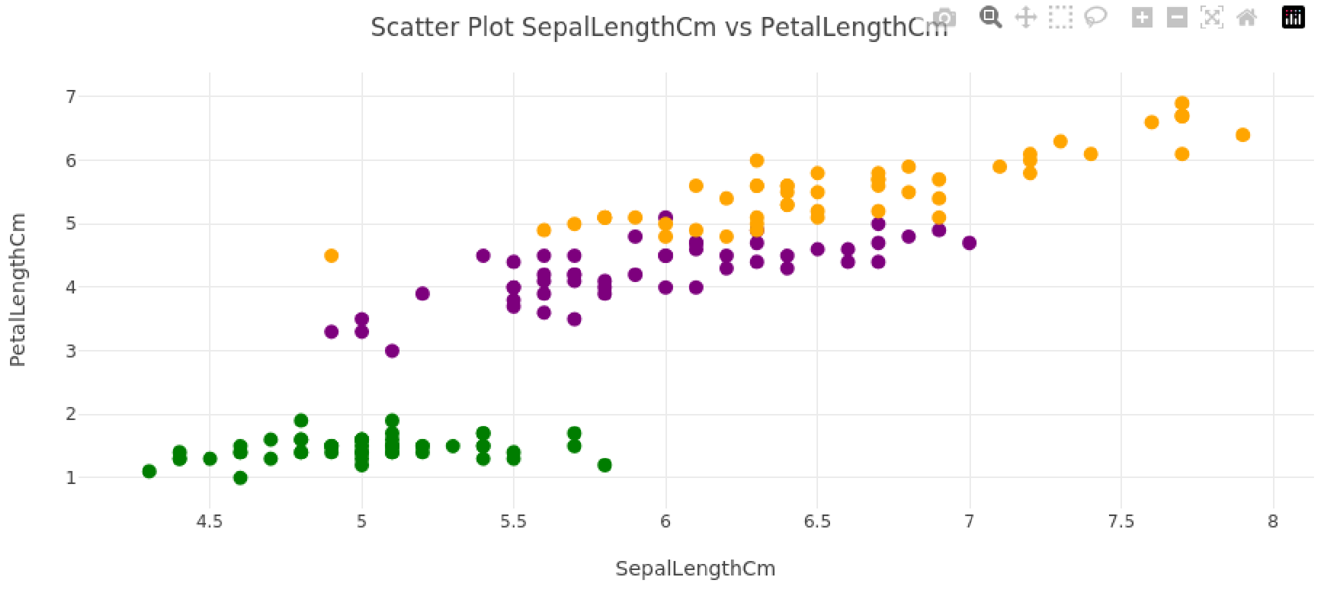
```
Pearson Correlation SepalLengthCm x PetalLengthCm: [[     1 0.8718]
 [0.8718 1.0000]]
```

Out[16]: 1

In [17]:
```
$trace = new Chart::Plotly::Trace::Scatter(
    x      => $X0->aspdl,
    y      => $X2->aspdl,
    mode   => 'markers',
    marker => {
        color      => $y->aspdl,
        colorscale => $color_scale,
        cmin       => $y->min,
        cmax       => $y->max,
        size       => 10
    }
);

$layout = {title => { text => sprintf('Scatter Plot %s vs %s', @$header[0, 2]) },
           xaxis => { title => $header->[0] }, yaxis => { title => $header->[2] },
           width  => 900, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                layout => $layout);
```

```
IPerl->display($plot);
```

In [18]:
```
sml->embed_plot($plot, width=>900, height=>450);
```



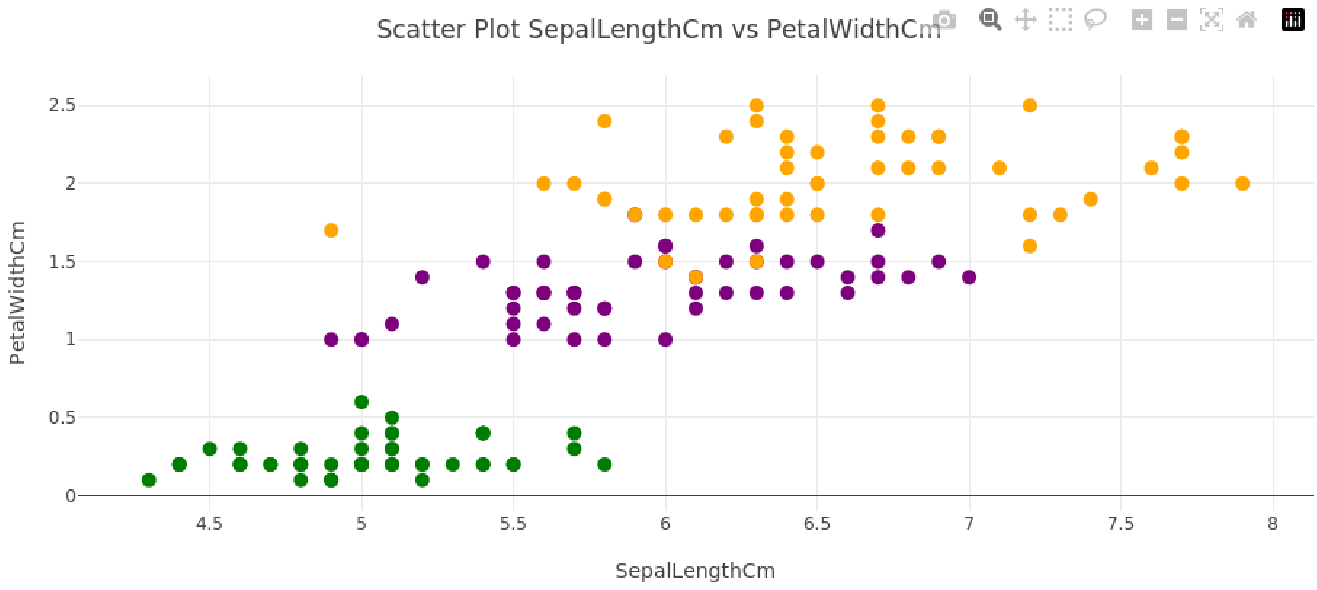
In [19]:
```
printf "Pearson Correlation %s x %s: %s\n", @$header[0, 3],
       nd->corrcoef($X0, y=>$X3)->asstr;
# Pearson Correlation X0 x X3
#  [[1.         0.81795363]
#  [0.81795363 1.        ]]
```

```
Pearson Correlation SepalLengthCm x PetalWidthCm: [[      1 0.8180]
 [0.8180      1]]
```

Out[19]: 1

In [20]:
```
$trace = new Chart::Plotly::Trace::Scatter(
    x      => $X0->aspdl,
    y      => $X3->aspdl,
    mode   => 'markers',
    marker => {
        color      => $y->aspdl,
        colorscale => $color_scale,
        cmin       => $y->min,
        cmax       => $y->max,
        size       => 10
    }
);

$layout = {title => { text => sprintf('Scatter Plot %s vs %s', @$header[0, 3]) },
           xaxis => { title => $header->[0] }, yaxis => { title => $header->[3] },
           width  => 900, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                layout => $layout);

IPerl->display($plot);
```

In [21]:
```
sml->embed_plot($plot, width=>900, height=>450);
```



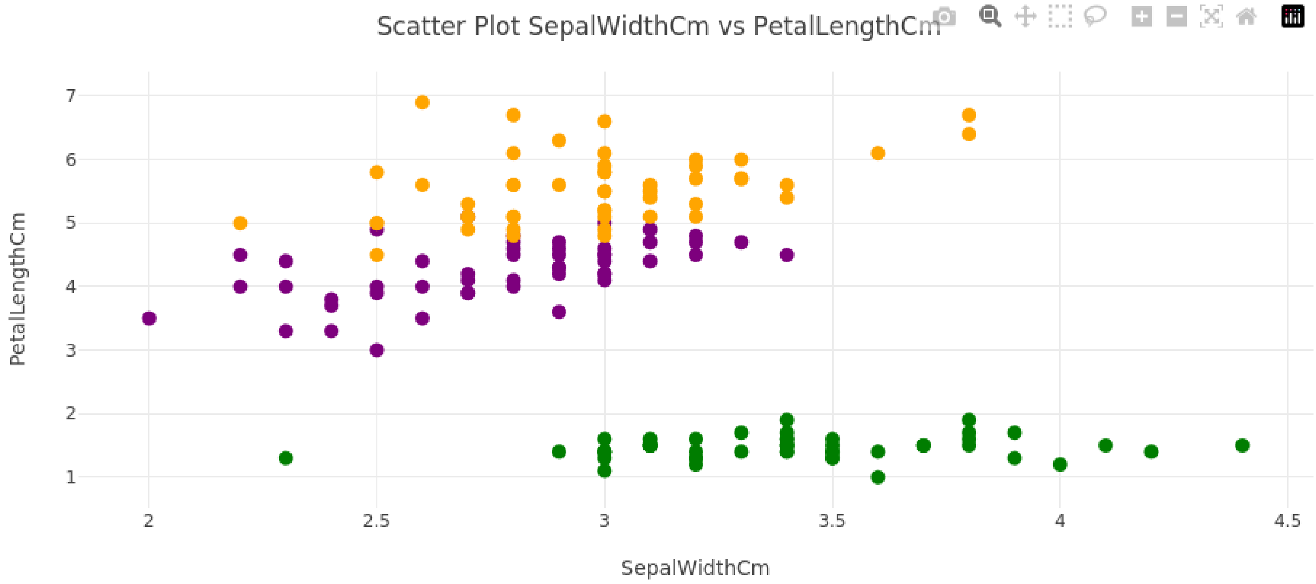
In [22]:
```
printf "Pearson Correlation %s x %s: %s\n", @$header[1, 2],
       nd->corrcoef($X1, y=>$X2)->asstr;
# Pearson Correlation X1 x X2
#  [[ 1.        -0.4205161]
#  [-0.4205161  1.       ]]
```

```
Pearson Correlation SepalWidthCm x PetalLengthCm: [[      1 -0.4205]
 [-0.4205  1.0000]]
```

Out[22]: 1

In [23]:
```
$trace = new Chart::Plotly::Trace::Scatter(
    x      => $X1->aspdl,
    y      => $X2->aspdl,
    mode   => 'markers',
    marker => {
        color      => $y->aspdl,
        colorscale => $color_scale,
        cmin       => $y->min,
        cmax       => $y->max,
        size       => 10
    }
);

$layout = {title => { text => sprintf('Scatter Plot %s vs %s', @$header[1, 2]) },
           xaxis => { title => $header->[1] }, yaxis => { title => $header->[2] },
           width  => 900, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                 layout => $layout);

IPerl->display($plot);
```

In [24]:
```
sml->embed_plot($plot, width=>900, height=>450);
```



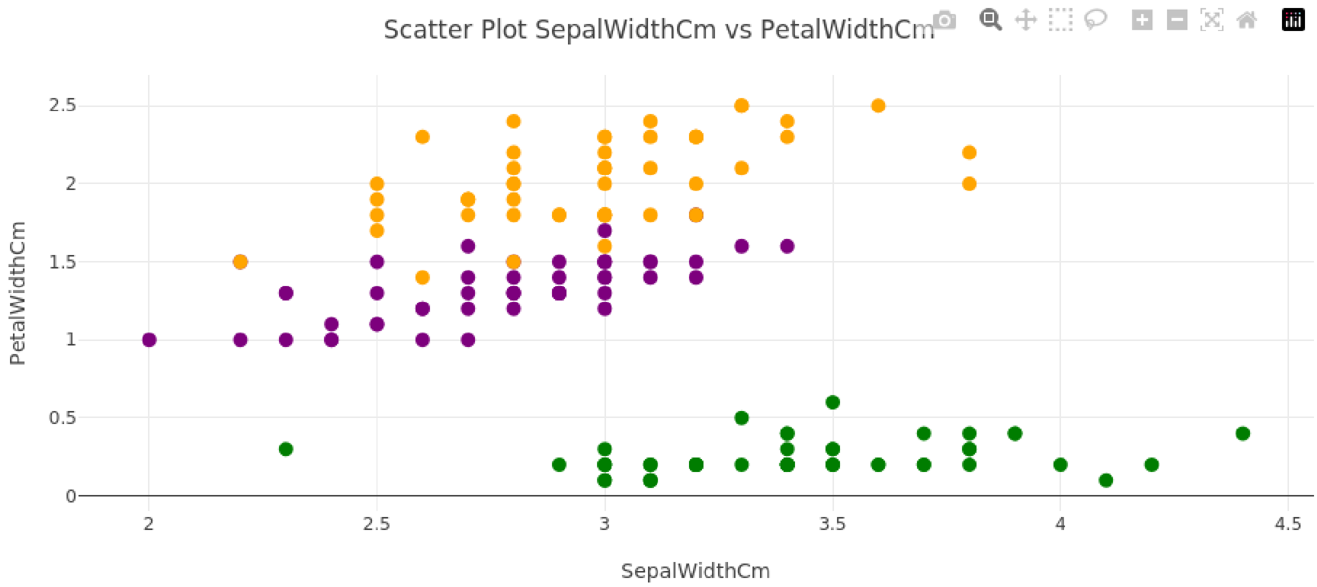
In [25]:
```
printf "Pearson Correlation %s x %s: %s\n", @$header[1, 3],
       nd->corrcoef($X1, y=>$X3)->asstr;
# Pearson Correlation X1 x X3
#  [[ 1.         -0.35654409]
#  [-0.35654409  1.         ]]
```

```
Pearson Correlation SepalWidthCm x PetalWidthCm: [[      1 -0.3565]
 [-0.3565       1]]
```

Out[25]: 1

In [26]:
```
$trace = new Chart::Plotly::Trace::Scatter(
    x      => $X1->aspdl,
    y      => $X3->aspdl,
    mode   => 'markers',
    marker => {
        color      => $y->aspdl,
        colorscale => $color_scale,
        cmin       => $y->min,
        cmax       => $y->max,
        size       => 10
    }
);

$layout = {title => { text => sprintf('Scatter Plot %s vs %s', @$header[1, 3]) },
           xaxis => { title => $header->[1] }, yaxis => { title => $header->[3] },
           width  => 900, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                layout => $layout);

IPerl->display($plot);
```

In [27]:
```
sml->embed_plot($plot, width=>900, height=>450);
```



In [28]:

```
printf "Pearson Correlation %s x %s: %s\n", @$header[2, 3],
       nd->corrcoef($X2, y=>$X3)->asstr;
# Pearson Correlation X2 x X3
#  [[1.         0.9627571]
#  [0.9627571 1.        ]]
```

```
Pearson Correlation PetalLengthCm x PetalWidthCm: [[1.0000 0.9628]
 [0.9628      1]]
```

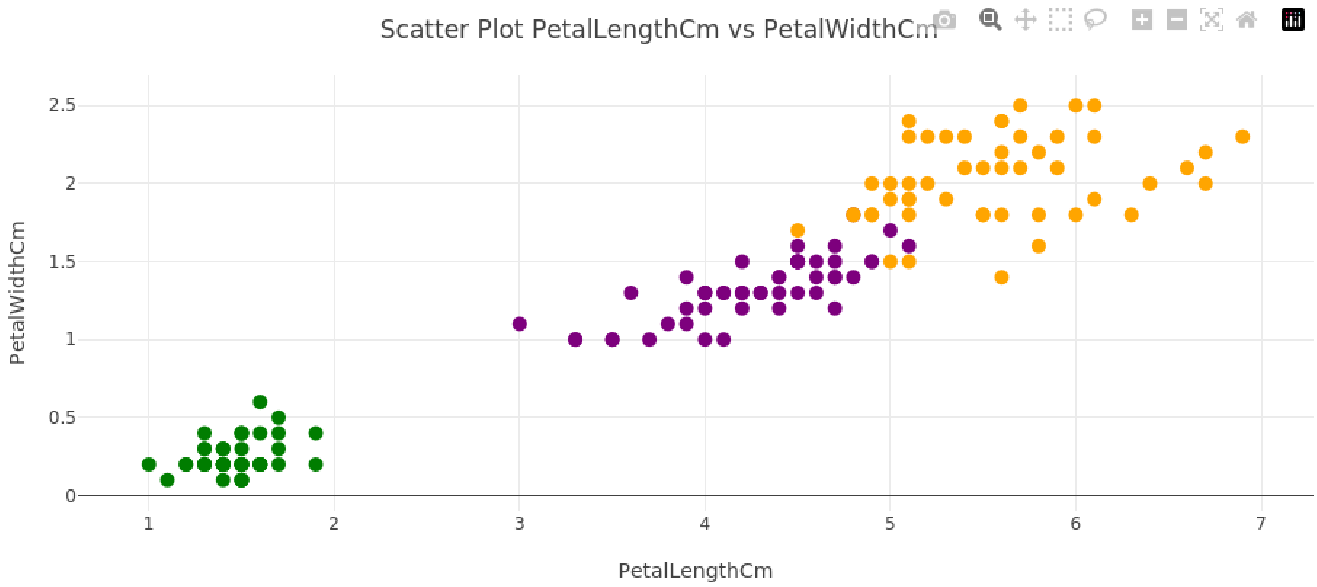
Out[28]: 1

In [29]:

```
$trace = new Chart::Plotly::Trace::Scatter(
    x      => $X2->aspdl,
    y      => $X3->aspdl,
    mode   => 'markers',
    marker => {
        color      => $y->aspdl,
        colorscale => $color_scale,
        cmin       => $y->min,
        cmax       => $y->max,
        size       => 10
    }
);

$layout = {title => { text => sprintf('Scatter Plot %s vs %s', @$header[2, 3]) },
           xaxis => { title => $header->[2] }, yaxis => { title => $header->[3] },
           width  => 900, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                layout => $layout);

IPerl->display($plot);
```

In [30]:

```
sml->embed_plot($plot, width=>900, height=>450);
```



In [31]:
```
my $X_normalized = $dataset->slice(':', [0, -1]);
my $y = $dataset->slice(':', -1);
# normalize
my $minmax = sml->dataset_minmax($X_normalized);
sml->normalize_dataset($X_normalized, $minmax);
my $normalized = nd->concat($X_normalized, $y->expand_dims(axis=>1));
nd->print("Normalized:", $normalized->slice([0, 5]));
```

```
Normalized:[[0.2222 0.6250 0.0678 0.0417      0]
 [0.1667 0.4167 0.0678 0.0417      0]
 [0.1111 0.5000 0.0508 0.0417      0]
 [0.0833 0.4583 0.0847 0.0417      0]
 [0.1944 0.6667 0.0678 0.0417      0]]
<AI::MXNet::NDArray 5x5 @cpu(0)> AI::MXNet::NDArray::Slice float32
```

Out[31]: 1

In [32]:
```
my $corrcoef_normalized = nd->corrcoef($normalized->T);
printf "Pearson Correlation coefficients of the Iris dataset: %s\n",
        $corrcoef_normalized->asstr;
```

```
Pearson Correlation coefficients of the Iris dataset: [[      1 -0.1094  0.8718  0.818
0  0.7826]
 [-0.1094  1.0000 -0.4205 -0.3565 -0.4194]
 [ 0.8718 -0.4205       1  0.9628  0.9490]
 [ 0.8180 -0.3565  0.9628  1.0000  0.9565]
 [ 0.7826 -0.4194  0.9490  0.9565       1]]
```

Out[32]: 1

In [33]:
```
printf "Pearson Correlation coefficients of X: %s\n", nd->corrcoef($dataset->T)->asst
# Pearson Correlation coefficients of X
#  [[ 1.         -0.10936925  0.87175416  0.81795363]
#  [-0.10936925  1.         -0.4205161  -0.35654409]
#  [ 0.87175416 -0.4205161   1.          0.9627571 ]
#  [ 0.81795363 -0.35654409  0.9627571   1.         ]]
```

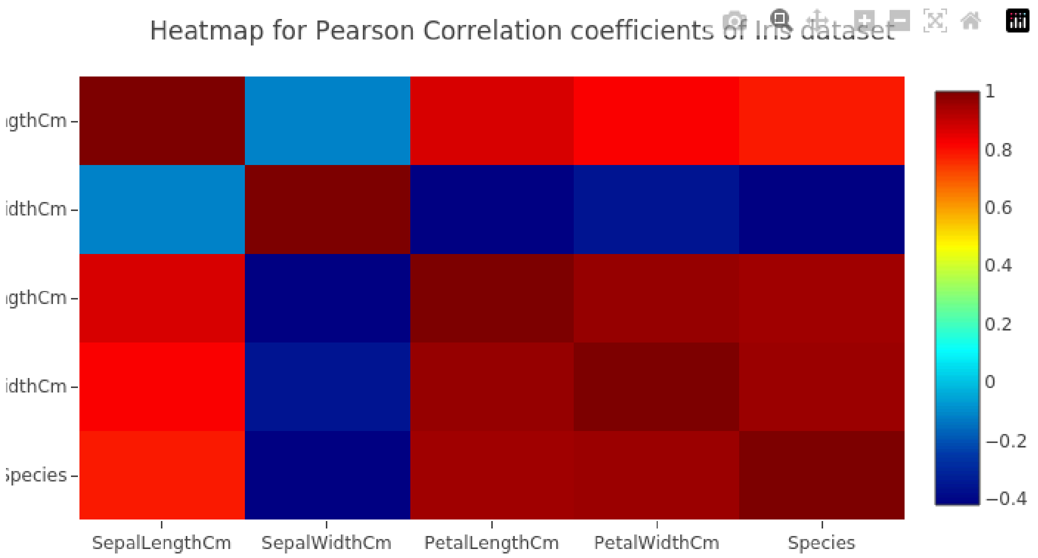
```
Pearson Correlation coefficients of X: [[      1 -0.1094  0.8718  0.8180  0.7826]
 [-0.1094  1.0000 -0.4205 -0.3565 -0.4194]
 [ 0.8718 -0.4205       1  0.9628  0.9490]
 [ 0.8180 -0.3565  0.9628  1.0000  0.9565]
 [ 0.7826 -0.4194  0.9490  0.9565       1]]
```

Out[33]: 1

In [34]:
```
# Show a heatmap out of the Pearson Correlation coefficients of X
$trace = new Chart::Plotly::Trace::Heatmap(
```

```
    x => $header,
    y => $header,
    z => $corrcoef_normalized->aspdl,
    colorscale => 'Jet', # Electric Greens Greys Hot Jet Picnic Portland Reds YlOrRd
                         # https://plotly.com/javascript/colorscales/#hot-colorscale
);

$layout = {title => { text => 'Heatmap for Pearson Correlation coefficients of Iris d
           yaxis => { autorange => "reversed" }, # Flip so lower frequencies are at t
           width  => 700, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                layout => $layout);

IPerl->display($plot);
```

In [35]:
```
sml->embed_plot($plot, width=>900, height=>450);
```



Looking at pima-indians-diabetes dataset

In [36]:
```
# load and prepare data
my $filename = '../data/pima-indians-diabetes.csv';
($dataset, $header) = sml->load_csv($filename, asndarray=>1);

my $X_normalized = $dataset->slice(':', [0, -1]);
my $y = $dataset->slice(':', -1);
# normalize
my $minmax = sml->dataset_minmax($X_normalized);
sml->normalize_dataset($X_normalized, $minmax);
my $normalized = nd->concat($X_normalized, $y->expand_dims(axis=>1));
printf "Normalized:%s", $normalized->asstr;
```

```
Normalized:[[0.3529 0.7437 0.5902 ... 0.2344 0.4833      1]
 [0.0588 0.4271 0.5410 ... 0.1166 0.1667      0]
 [0.4706 0.9196 0.5246 ... 0.2536 0.1833      1]
 ...
 [0.2941 0.6080 0.5902 ... 0.0713 0.1500      0]
 [0.0588 0.6332 0.4918 ... 0.1157 0.4333      1]
 [0.0588 0.4673 0.5738 ... 0.1012 0.0333      0]]
```

Out[36]: 1

In [37]:
```
my $corrcoef_normalized = nd->corrcoef($normalized->T);
printf "Pearson Correlation coefficients of the Diabetes dataset: %s\n",
        $corrcoef_normalized->asstr;
```

```
Pearson Correlation coefficients of the Diabetes dataset: [[      1  0.1295  0.1413 -
0.0817 -0.0735  0.0177 -0.0335  0.5443  0.2219]
 [ 0.1295       1  0.1526  0.0573  0.3314  0.2211  0.1373  0.2635  0.4666]
 [ 0.1413  0.1526       1  0.2074  0.0889  0.2818  0.0413  0.2395  0.0651]
 [-0.0817  0.0573  0.2074  1.0000  0.4368  0.3926  0.1839 -0.1140  0.0748]
 [-0.0735  0.3314  0.0889  0.4368       1  0.1979  0.1851 -0.0422  0.1305]
 [ 0.0177  0.2211  0.2818  0.3926  0.1979       1  0.1406  0.0362  0.2927]
 [-0.0335  0.1373  0.0413  0.1839  0.1851  0.1406       1  0.0336  0.1738]
 [ 0.5443  0.2635  0.2395 -0.1140 -0.0422  0.0362  0.0336       1  0.2384]
 [ 0.2219  0.4666  0.0651  0.0748  0.1305  0.2927  0.1738  0.2384       1]]
```

Out[37]: 1

In [38]:
```
# standardize
my $X_standardized = $dataset->slice(':', [0, -1]);
# Estimate mean and standard deviation
my $means = sml->column_means($X_standardized);
my $stdevs   = sml->column_stdevs($X_standardized, $means);

# standardize dataset
sml->standardize_dataset($X_standardized, $means, $stdevs);
my $standardized = nd->concat($X_standardized, $y->expand_dims(axis=>1));
print $standardized->asstr;
```

```
[[ 0.6395  0.8478  0.1495 ...  0.4682  1.4251       1]
 [-0.8443 -1.1227 -0.1604 ... -0.3648 -0.1905       0]
 [ 1.2331  1.9425 -0.2638 ...  0.6040 -0.1055       1]
 ...
 [ 0.3428  0.0033  0.1495 ... -0.6847 -0.2756       0]
 [-0.8443  0.1597 -0.4704 ... -0.3709  1.1700       1]
 [-0.8443 -0.8725  0.0462 ... -0.4735 -0.8708       0]]
```

Out[38]: 1

In [39]:
```
my $corrcoef_standardized = nd->corrcoef($standardized->T);
printf "Pearson Correlation coefficients of the Diabetes dataset: %s\n",
        $corrcoef_standardized->asstr;
```

```
Pearson Correlation coefficients of the Diabetes dataset: [[ 1.0000  0.1295  0.1413 -
0.0817 -0.0735  0.0177 -0.0335  0.5443  0.2219]
 [ 0.1295       1  0.1526  0.0573  0.3314  0.2211  0.1373  0.2635  0.4666]
 [ 0.1413  0.1526       1  0.2074  0.0889  0.2818  0.0413  0.2395  0.0651]
 [-0.0817  0.0573  0.2074       1  0.4368  0.3926  0.1839 -0.1140  0.0748]
 [-0.0735  0.3314  0.0889  0.4368       1  0.1979  0.1851 -0.0422  0.1305]
 [ 0.0177  0.2211  0.2818  0.3926  0.1979  1.0000  0.1406  0.0362  0.2927]
 [-0.0335  0.1373  0.0413  0.1839  0.1851  0.1406       1  0.0336  0.1738]
 [ 0.5443  0.2635  0.2395 -0.1140 -0.0422  0.0362  0.0336  1.0000  0.2384]
 [ 0.2219  0.4666  0.0651  0.0748  0.1305  0.2927  0.1738  0.2384       1]]
```

Out[39]: 1

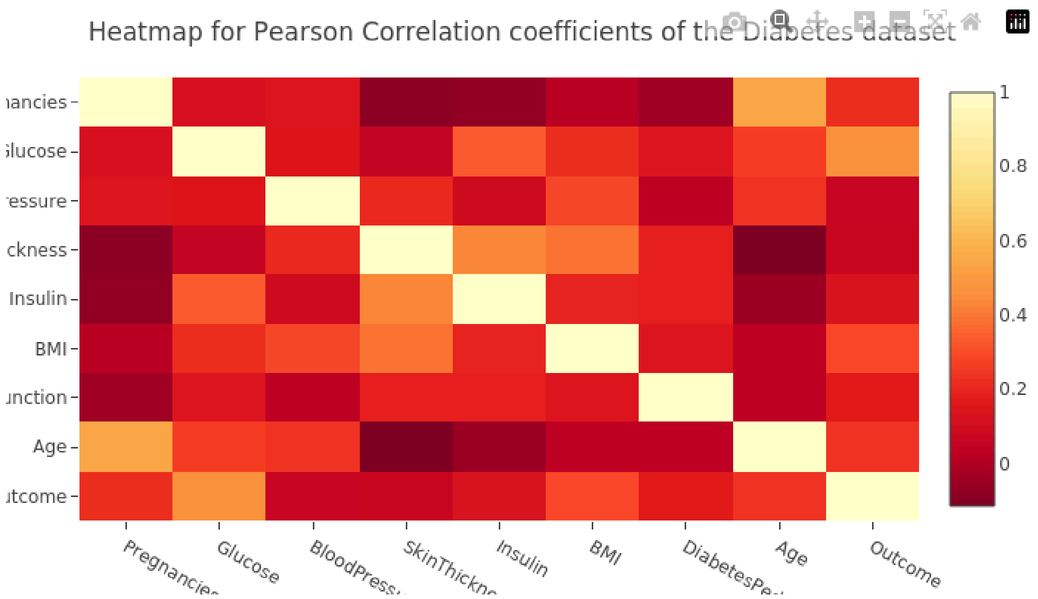
In [40]:
```
printf "%d-%s ", $_, $header->[$_] for (0 .. $#$header);
```

```
0-Pregnancies 1-Glucose 2-BloodPressure 3-SkinThickness 4-Insulin 5-BMI 6-DiabetesPedi
greeFunction 7-Age 8-Outcome
```

In [41]:
```
# Show a heatmap out of the Pearson Correlation coefficients of X
my $trace = new Chart::Plotly::Trace::Heatmap(
    x => $header,
    y => $header,
    z => nd->corrcoef($normalized->T)->aspdl,
    colorscale => 'YlOrRd', # Electric Greens Greys Hot Jet Picnic Portland Reds YlOr
                            # https://plotly.com/javascript/colorscales/#hot-colorsca
```

```
);

$layout = {title => { text => 'Heatmap for Pearson Correlation coefficients of the Di
           yaxis => { autorange => "reversed" }, # Flip so lower frequencies are at t
           width  => 700, height => 400,
           margin => { l => 50, r => 0, t => 50, b => 50 }};

$plot = new Chart::Plotly::Plot(traces => [$trace],
                                layout => $layout);

IPerl->display($plot);
```

In [42]:
```
sml->embed_plot($plot, width=>900, height=>450);
```



In [ ]: